

Protocol Specification

Web ↔ Local Engine (Production)

1. Scope

Defines the bidirectional, versioned protocol between the web client and the local audio engine over localhost WebSocket.

Transport: WebSocket (ws://127.0.0.1:{PORT}) Encoding: JSON (UTF-8) State: session-oriented, multiplexed by sessionId

2. Versioning

- protocolVersion: integer (required in every message)
- Backward compatibility: N supports N-1 for non-breaking fields
- Breaking changes require version bump and capability negotiation

Handshake requires version match or compatible range.

3. Envelope

All messages MUST follow:

```
{
  "type": "string",
  "protocolVersion": 1,
  "sessionId": "string|null",
  "timestamp": 0,
  "requestId": "string|null",
  "payload": {}
}
```

Rules: - timestamp: ms since epoch (engine time for engine→web, client time for web→engine) - requestId: required for request/response pairs - sessionId: required for session-scoped messages

4. Connection Lifecycle

4.1 Handshake

Web → Engine:

```
{ "type": "init", "protocolVersion": 1, "requestId": "r1", "payload": {  
  "client": "web", "minVersion": 1, "maxVersion": 1 } }
```

Engine → Web:

```
{ "type": "ready", "protocolVersion": 1, "requestId": "r1", "payload": {  
  "engineVersion": "x.y.z", "protocolVersion": 1, "capabilities": ["audio",  
    "scoring"] } }
```

4.2 Heartbeat

- Interval: 5s
- If 3 consecutive heartbeats missed → connection lost

Engine → Web:

```
{ "type": "heartbeat", "protocolVersion": 1, "payload": { "uptimeMs":  
  123456 } }
```

5. Device Management

5.1 List Devices

Web → Engine:

```
{ "type": "get_devices", "requestId": "r2" }
```

Engine → Web:

```
{ "type": "devices", "requestId": "r2", "payload": { "devices": [{ "id":  
  "d1", "name": "Scarlett", "sampleRates": [44100, 48000] } ] } }
```

5.2 Select Device

```
{ "type": "select_device", "payload": { "deviceId": "d1", "sampleRate":  
  44100, "bufferSize": 256 } }
```

Ack:

```
{ "type": "device_selected", "payload": { "deviceId": "d1" } }
```

6. Calibration

```
{ "type": "calibrate_start" }
```

```
{ "type": "calibration_result", "payload": { "inputLevel": -12.3,  
"recommendedThreshold": 0.2, "latencyMs": 28 } }
```

7. Session Control

7.1 Start

```
{ "type": "start_session", "sessionId": "s1", "payload": { "bpm": 120,  
"noteMap": [...], "windows": { "perfect": 30, "good": 80 } } }
```

Engine → Web:

```
{ "type": "session_started", "sessionId": "s1", "payload": {  
"engineStartTime": 0 } }
```

7.2 Pause/Resume/Stop

```
{ "type": "pause_session", "sessionId": "s1" }  
{ "type": "resume_session", "sessionId": "s1" }  
{ "type": "stop_session", "sessionId": "s1" }
```

8. Realtime Events

8.1 Note Event

```
{ "type": "note_event", "sessionId": "s1", "payload": { "pitchHz": 329.63,  
"note": "E4", "confidence": 0.92, "onsetMs": 3245, "timingOffsetMs": -18,  
"judgement": "good" } }
```

8.2 Score Update

```
{ "type": "score_update", "sessionId": "s1", "payload": { "combo": 7,
"score": 1230, "accuracy": 0.87 } }
```

8.3 Session Result

```
{ "type": "session_result", "sessionId": "s1", "payload": { "score": 3450,
"accuracy": 0.91, "hits": 120, "misses": 15, "engineVersion": "x.y.z" } }
```

9. Errors

Standardized errors: - ENGINE_NOT_READY - DEVICE_NOT_FOUND - AUDIO_INIT_FAILED - INVALID_SESSION - VERSION_MISMATCH

```
{ "type": "error", "payload": { "code": "DEVICE_NOT_FOUND", "message":
"..."} }
```

10. Security

- Only allow localhost origin
 - Optional session token in init payload
 - Reject unknown origins
 - Validate all payloads (schema)
-

11. Guarantees

- Engine is source of truth for scoring
 - All realtime events are ordered per session
 - Idempotent handling for control messages
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12. Definition of Done

Protocol is complete when: - All message types implemented - Version negotiation works - Reconnect and heartbeat stable - End-to-end session runs without schema violations